Complex Technical Document

1. System Overview
The platform coordinates document loading, semantic chunking, and hybrid indexing.
It keeps ingestion deterministic while exposing collection metadata for later retrieval.

2. Data Contracts
Every chunk carries source_path, collection, chunk_index, and optional image_refs.
A stable SHA256 document identifier links vector, sparse, and image artifacts together.

Table 1: Stage Contract

Stage	Input	Output
load	PDF path	Document
split	Document	Chunk[]
store	Records	Index files

3. Retrieval Strategy
Dense embeddings capture semantic similarity while BM25 protects exact-match recall.
A lightweight rerank step can refine the final candidate set when latency budget allows it.

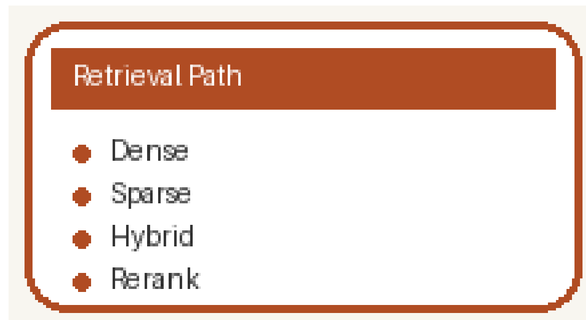
4. Failure Handling
Each pipeline stage wraps lower-level errors with a readable stage prefix for operators.
Failures still mark file integrity status so repeated runs can be diagnosed quickly.

Table 2: Retrieval Modes

Mode	Strength	Tradeoff
dense	semantic recall	embedding cost
sparse	keyword precision	lexical only

Table 3: Error Policy

Condition	Action
loader exception	mark failed
duplicate file	skip unless forced

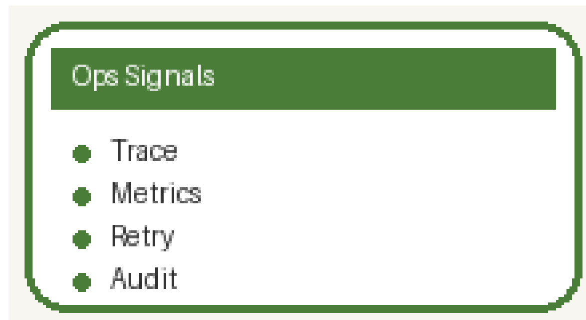


5. Multi-Modal Enrichment
Image placeholders preserve the location of diagrams so later captioning can fill in context.
Metadata enrichment may append page-level hints, titles, and structured tags.

6. Batch Encoding
Dense and sparse encoders run over the same chunk list and return aligned record identifiers.
Batch size is configurable from settings to balance throughput and external API rate limits.

Table 4: Encoding Outputs

Encoder	Output
dense	float vector
sparse	token weight map



7. Storage Layout
Image bytes persist under collection-scoped folders while SQLite stores image_id mappings.
BM25 files live beside vector artifacts so offline ingest remains reproducible.

8. Operations Checklist
Before running ingest, confirm provider credentials, collection names, and source paths.
After completion, validate chunk count, image count, and index presence in storage backends.

Table 5: Runbook

Check	Expected
logs	stage success messages
images	extracted files exist
indexes	vector and BM25 artifacts exist